Wharton Sports Analytics and Business Initiative

Wharton High School Data Science Competition
Webinar 2
February 19, 2026 9:00AM EST

Jamie Powers: Hi, everyone, and welcome. I'm Jamie Powers, the Program and Operations Specialist for the Wharton Sports Analytics and Business Initiative. I'm joining you today from Philadelphia, Pennsylvania.

It's great to have you here today for today's webinar, and I want to start by thanking you for participating in the third year of the Wharton High School Data Science Competition.

This competition is an incredible opportunity for you to apply your data science skills in a real-world context, and we're excited to see the innovative work you've been doing.

Today, we're fortunate to have two data scientists from the competition team joining us today to answer your questions. Please welcome Liz Beard and Russ Walters.

Over the next 60 minutes, we'll focus on answering your pre-submitted questions. We've received a lot of great ones, and we'll get through as many as we can. If you think of any additional questions during the webinar, feel free to type them into the Q&A box at the bottom of your screen. Be sure to include your team's name when you submit the question. While we may not get to all of them, we'll do our best to cover as much as possible.

Before we get started, I'm gonna do a few housekeeping items.

So to start off, if you have any team photos, we'd love to see them. So you can send them to ai-analytics@warton.upenn.edu. These are due by February 25th. You can find submission instructions in last week's message.

Number two, as a reminder, your team's student leader has to mark the competition prompt as complete in order to advance, and that has to be done by tomorrow, February 20th, at 9 am Eastern Time. So, for the few teams who have not completed this task, please do so now.

All teams who have not marked the task complete by the deadline will be eliminated.

And then finally, Phase 1 submissions are due by March 2nd. So if you haven't already, please make sure you're on track to submit by then.

Alright, so we can get started and jump into some questions. We have these grouped by category, so we're going to go category by category. We're gonna start off with some data questions.

So, the first one is from Outliers Analytics, and they asked.

For some records in the primary data, the time spent on ice is significantly smaller than the penalty time minutes. Does this mean that

the penalty time for the penalties committed in Record 1 may be passed down or carried to the following records as well?

Russel Walters: In the dataset that you have, the time on ice is the game summary for each of the line-pair combinations.

If one line commits a penalty, then the penalty unit comes on the ice. So, a penalty that's committed by one line results in a power play, and the power play is the power kill pair that comes on afterwards. So yeah, this is not surprising, this is the way that the dataset will work, where one combination of lines has a penalty against it, and then the power play unit comes on the ice after that.

Jamie Powers: Awesome, thank you. So, for the second question, it's from Arcadia Titans.

Does the top 10 in offensive line quality disparity refer to the teams with the biggest gap in quality, or the smallest gap in quality between offensive lines?

Liz Beard: Awesome, I'm happy to answer that one. So, for the way that we've been conceptualizing it, the offensive line quality should start with the teams from the largest gap to the smallest gap, or we found it easier to think about it in terms of the ratio, so… The larger the ratio, or the more disparity, would be the higher the teams in offensive line quality disparity.

Jamie Powers: Sounds good. Alright, the next question is from Cougar Troopers.

Is there a timeline for the dataset? By which I mean, is the data an order of which games happen first, and which one occurs after that? Or should it be assumed that each event occurs individually with no regard to specific order?

Russel Walters: I the educational modules, there's no time information in the dataset, there's no date sequence, so the games do have a game ID, `game_` 1, 2, 3, counting up sequentially, but that sequence doesn't mean anything.

Keep in mind that the team quality doesn't change over the course of the season, the team composition doesn't change, and the players don't change.

So this is a significant simplification from actual hockey. In our league, there's no change over time, no change over the course of the season.

Jamie Powers: Sounds good. Alright, the next question from Apex Predators.

Our team is seeking clarification on the overtime procedures. As we've found conflicting information in the official materials, the

instructional video mentions a sudden death format, whereas the glossary documents suggest a 5-minute period followed by a shootout.

It's a two-part question, if you want to answer that one first.

Russel Walters: The glossary over time definition was general. I've updated it to make it specific to our league. So, in our league, overtime continues in the same manner as regular play, so at the end of regulation, if the game's tied, teams continue playing until a game is scored, however long that takes. There's one game, I think, that runs for an extra, like, 45 minutes or something for a long time.

But overtime play is the same as regular play, also a simplification from NHL hockey. The teams keep playing in the same manner until a game is scored.

Jamie Powers: Alright, and then the second part of the question is, additionally, we've observed games entering overtime despite not being tied at the 60-minute mark. As well as games where, multiple points were scored during the overtime period. Can you please confirm the exact protocol for when overtime is triggered and how it is played?

Russel Walters: The dataset you have has game summary data for each line pair combination, and that summary is over the entire game, regulation, and over time, so you can't actually tell what the score was at a given time.

If the game goes to overtime, it was tied, and then the additional goal happens in overtime. Again, you don't have play-by-play data, you don't know when the goals were scored, you just know what line combinations scored the goals.

Jamie Powers: Next question is from Epoch Learn.

Can you explain the empty net record a bit more? In real hockey, the empty net usually happens at the end of the game as a desperate mood move, and is not very predictive for team skills. Is this the case for the synthetic data?

Russel Walters: Yes, the empty net, also described in the educational models, the empty net acts like NHL hockey. As mentioned, at the end of the game, it occurs if one team's down, and they're trying to tie the game. So, you know, consider the team that has the empty net, if they have a goal scored against them, that probably does not reflect much in team quality.

But the team that has the man advantage, who scores a goal, if they do, with the goalie in place, that probably has more information than the opposite case.

Jamie Powers: Alright, the next question is from Arcadia Titans.

For round one of the playoffs, is it one game or a 7-game series where the home advantage flips?

Liz Beard: That's a single game.

Jamie Powers: Alright. Outliers Analytics asked, for the home team win probability task, should probabilities reflect the final match outcome, including overtime and shootout, or just the regulation time?

Russel Walters: There's no shootout in our league, play continues in the same manner. So, yeah, everything functions the same if there's overtime or not… consider a regulation game.

Jamie Powers: Flatbush Falcons would like to know, can we use the data that wasn't specified in the resources, by using outside information?

Russel Walters: You can use whatever you like, but we'd caution partly against that. The information that you have is sufficient to solve the case, but I mean, certainly, if you want to use methodologies that you learn outside, but bringing outside data into the case is probably not recommended.

Jamie Powers: Chickenhockey asked, is it reasonable to use ELO ratings when no date or sequence information is given?

Russel Walters: We're not going to talk too much about methodology choices, mostly we'll be clarifying data issues. So, I'm not going to answer that directly. You can do lots of different things, lots of different methodologies make sense. Using ELOs is a fine method, so I'm not going to talk extensively about methodology choices.

That is, aside from what's in the educational modules.

Jamie Powers: Sounds good. I just want to jump in really quickly. If you guys have questions, you can drop them into the Q&A box. I see some people raising their hands, so I just want to let you know that.

Jamie Powers: Alright, so Data Dragons asks, can we assume that shot minus goals equals saves? As in a real hockey match, there could be some shots which were just way off and rebounded or never came near the goal.

Russel Walters: Yeah, it's also handled in the educational modules, but yes, shots minus goals is saves.

Jamie Powers: Alright, and then Data Dragons also asked, in the simulated data, do the players get tired as the game progresses, or can we assume their performance is consistent in that aspect?

Russel Walters: Yeah, in the simulated data, like in raw hockey, players get tired, and that happens in the simulation, though, it's a little bit hard to see, because you don't have the time sequence of the in-game activity, but different teams can differ.

Jamie Powers: Alright, the next two questions are from Spartan Analytics Club.

So the first one is, what does it mean when there are non-power play roles? But with home penalty minutes or away penalty minutes, shouldn't those penalty minutes be accounted for in the power play rows instead?

Russel Walters: Like I mentioned before, the data is game summary data, across the line payer combinations. So, if the top line against the other top line commits a penalty, that penalty results in a power play, which is a different unit on the ice next. Power play units don't typically create, or commit a lot of penalties; most of the penalties are committed by the full-strength line-pair combinations, but then the power play happens with the power play line.

Jamie Powers: Alright, and then their next question is, which team dictates the line matchups? Does the home or away teams send out their lines first?

Russel Walters: Yeah, again, you can't really see this in the Advantage Center, it's rolled up and summarized across all the pairings, and as you can see, typically, most of the pairings occur in a game, so top line plays second line, second line plays top line, across all of the possible pairings, and as you can see in the data set, the different line combinations and defensive pairings vary in time on ice together. But most of the different combinations occur across the course of the game. Then you have the game summary results of those.

Jamie Powers: Alright, and then for our pre-submitted questions, this is the last data category one.

The SEALs would like to know, when calculating offensive line quality disparity, should we adjust expected goals for time on ice and defensive matchup strength? Or is using raw XG values acceptable for ranking teams?

Russel Walters: So, again, I'm not going to talk about methodology too much, but the first part of that sounds like a good idea.

Jamie Powers: So, we're gonna move on to some questions about grading and the finals.

We have a lot of similar questions, so if you asked a question about this It's going be the same answer all around, so I'll just read one of the top ones that came in, the cat that overfit asked,

What metric or match-up predictions and team rankings graded on? How are the semifinalists chosen?

Liz Beard: Awesome. I'm happy to try and take this one, hopefully my Wi-Fi is more stable now, sorry about that, everyone. But kind of similarly, like, we're not gonna talk too much about specific methodologies, I'm not going to go into Super great detail about how exactly our team determines the semi-finalists, and how we kind of grade everybody's submissions, but I encourage people to go back and look at the competition prompt again. Essentially, what teams are going to be scored on are things like the accuracy of their responses, the strength of the methodology that you use, and the clarity of communication of how you got to those answers,

and the methodologies that you used, so our scoring is generally a balance between those three factors, and because of that, there's not necessarily one right answer that anyone should be striving for. It's kind of about your approach as a whole.

Jamie Powers: Now, we have another question from Flatbush Falcons that asks, do teams only work on the second phase of the competition after being selected to continue? If so, how many teams are selected to continue to Phase 2?

Liz Beard: Yes, only teams that are selected as semifinalists after submitting their Phase 1 answers will continue on to Phase 2.

And Phase 2 will focus on, if you are selected for the semifinals, you'll get more information about this, but it will kind of dive deeper into sharing those Phase 1 approaches and communicating your results to a broader audience. We generally select around 25 teams for the semifinalists. And I think that's pretty much it.

Jamie Powers: Alright, so those were all of our grading questions. We're gonna move on to just a few miscellaneous ones that came through.

Mountain View Analytics wants to know, how exactly was the data simulated? Does Wharton have an underlying model that created the data?

Russel Walters: Yes.

Liz Beard: we have an underlying model. We're not going to reveal anything about it until after the competition is over, but because this is one of the first times we've really simulated the data in this way. We plan to talk more about it in the playbook that will be sent out to all teams that submit answers to Phase 1 at the end of the competition.

Jamie Powers: Alright, and then Kellett Mighty Pucks wants to know, how long should we take per phase on average?

Liz Beard: Yeah, so this is also something that I recommend people check out the FAQs on the High School Data Science Competition webpage outside of SurveyMonkey Apply. I know it's a little complicated with information being in multiple places, but we generally suggest it takes around 4 to 6 hours per week for each phase. And Phase 1 definitely takes the longest. It's kind of the bulk of the work that you'll be doing for the competition.

From a data science perspective, a lot of it afterwards is kind of more around science communication.

Jamie Powers: Now we're going to move on to questions about submissions and any SurveyMonkey Apply questions.

So, Epoch Learn wants to know. Can I edit the Phase 1A responses any time before the deadline, or is it fixed with when I click next?

Liz Beard: Mmm, important questions. You can edit your Phase 1 responses up until the final page of the submission form. And in big bold text, there's a section that says, once you click "mark as complete", your answers are submitted, you cannot edit them, but up until that time, if you want to save your responses, I recommend clicking the "save and exit" at the end of the page, whenever you're finished working on it.

So just look out for that final page. It will come after a series of feedback form questions. So, if you're going through the submission form, once you hit those, you know you're kind of nearing the end of the submission form.

Jamie Powers: Meraider analysts want to know, when entering WIN probabilities for Phase 1B, SurveyMonkey says that there is an error when I try to put decimals in my answer. I was also wondering how many places I should round the probabilities to.

Liz Beard: Awesome, thank you for bringing that up. So, we looked into this a little bit. We would like you to round to the second decimal place, or the hundredth, I believe. Anything more than two decimal places will create and generate an error.

Jamie Powers: Milton Eagles asked, typically, what have the projects been in the past years? When we submit our data, what format… okay, well, actually, this is a four-part question, so do you want to go one at a time?

Liz Beard: Yeah, sure, that sounds great.

Jamie Powers: So, first one, typically, what are the projects like in the past years.

Liz Beard: So in the past years, if you want to learn more about those projects, I definitely recommend checking out our webpage, because we do highlight some of the student teams, and the winners from years past. They're all going to be a little bit different, because for the past 3 years, we've done a different sport, and we've done some slightly different types of prediction problems.

And we also have our sports analytics journal, right, Russ? That this winning team can submit to, so if you really want to look and dig into it and their approach, I recommend checking that out also.

Jamie Powers: And that journal is on the Wharton Sports Analytics and Business Initiative website, so it's easy to find if you have looked at the competition page.

The second question was, when we submit our data, what format or type are we supposed to use?

Liz Beard: So, I don't necessarily think you'll be submitting any data, but what you will be submitting is those rankings and predictions. If you haven't yet, I encourage you to click through the Phase 1 prediction form, just to see what it looks like, but essentially, you'll be entering

in the predictions that your data and your models have generated, not the raw data itself.

Jamie Powers: If we use outside sources, how do we cite it?

Liz Beard: Great, so you can cite your methods in any way that is clear and concise. There are some word limits I know people need to account for. Generally, APA, the American Psychological Association Standard is a good one to use. They also have a short-form citation that can help you save on space if you're worried about word count.

And, if you use any outside sources, it's definitely important to cite them in the methods section.

Jamie Powers: Alright, and then the final question from the Milton Eagles was, if you had to solve it, what resources do you think would be the most helpful?

Liz Beard: I think the most helpful resources would be the ones that our team has worked on developing over the past, 6 months, maybe longer. So watching the educational modules that are available, reading the workbook, and looking at the past projects on our website can help, but because the prompt and the sport has changed every year, what approach worked for those teams may not work best for hockey or for this competition. But those are kind of the three main resources I would focus on, especially since this simulated dataset in the World Hockey League functions a little bit differently than traditional, real-world hockey.

Jamie Powers: Alright, and then… A-M-I-T-Y-V1 stars want to know, for the visualization, is there any preferred format or resolution?

Liz Beard: There's no format resolution requirements, your visualization should be legible and clear, and there is a 5MB file size limit, so as long as you're within those constraints, you should be good to upload them.

Jamie Powers: Alright, so that was all that we had pre-submitted. Again, there was a few similar questions coming in, so if you didn't hear your team name, it was probably answered.

Now we do have a few coming in in the chat that we're gonna move on to. So…

For Phase 1A, the prompt asks to predict the win probability for the home team For 16 matchups. How exactly are these probabilities scored for accuracy? Will the judges use a continuous metric? Or is it evaluated strictly on binary win-loss accuracy?

Russel Walters: It will be a continuous metric, so something like log loss.

Jamie Powers: Novi High School asked, What role should expected goals play in power rankings?

Russel Walters: I'm not going to answer too many methodology questions, educational modules that contain the expected goal information are probably useful to look at. There's more information in expected goals than goals. So, maybe consider those carefully.

Jamie Powers: Alright, HCIS datasets, asked what metric will be used to evaluate predictions?

Liz Beard: I know we touched on this a little bit earlier, and Russ just talked a little bit about how we're going to look specifically at the predicted probabilities, but, in general, we're taking a holistic approach to the competition, competition prompt, and the four phases 1A, B, C, and D of the competition.

Jamie Powers: Alright, next question. What exactly counts as predicted tournament matchup outcomes? Is it all rounds, or just the first round?

Liz Beard: You can think about it as the first round.

Jamie Powers: Bulldogs27 asked, offensive line disparity between the first and second line is not only if the first line is better, right? Like, if a first line is much better than the second line, and for a different team, the second line is much better than the first line, and say both those teams have the two largest disparities by absolute value, should I count both of those at the top?

Or just if the first line is better As the team with the most disparity and the best second line as the team with the least disparity. Sorry, that was a long one.

Russel Walters: The first line is typically better than the second line. It's probably true across most teams. Some teams will be similar. That's why it's the top line and then the second line. And the disparity is within the team, so Team A's top line compared to Team A's second line. So, we're looking for the disparity within team disparity between lines.

Jamie Powers: Next question asks, the prompt notes that there are no changes in team or line quality over the course of the season. Does this mean time series trends, like a team getting on a hot streak right before the playoffs can be safely ignored, allowing us to treat all 82 teams as independent events?

Russel Walters: Right. Yeah, the teams don't change. There's no hot streaks there, there's no composition changes, players don't get hurt in our wonderful league.

Jamie Powers: Phase 1C requires a single PNG file upload. Is it acceptable to include multiple subplots within the single image, or must it strictly be one singular graph?

Liz Beard: That is up to you as a team, but that being said, one of the things that we are judging the strength of the submissions on is clarity of communication, so it's really important for you to think about what

information is most important to convey, and if you can convey it concisely.

And just as a brief aside, I'd also like to mention, I see a couple people raising their hands. If you could put your questions in the Q&A, box with your team name, that would be really helpful. That will help us be able to get to your question.

Jamie Powers: Next question asks, could you please elaborate on what the submission is for 1C? Are we supposed to submit a paper or a set of figures and tables?

Liz Beard: No paper, don't submit a paper. Love that you're thinking about writing a paper, though, that's awesome. But for 1C, you're only supposed to, submit a data visualization, so it can be a single graph, it can be a multi-panel that contains multiple graphs. One question just asked about that and I just answered. But, no, please don't submit a full paper.

Jamie Powers: And that actually brings us to the next question, is who will grade our responses?

Russel Walters: Not going to be explicit about that, but you might be looking at them. Or most of us.

Liz Beard: Couple people on this call.

Jamie Powers: Phase 1B requires creating an expected goals measure that accounts for defensive matchups, since tougher opponents can affect performance. Are there specific defensive metrics within the provided data dictionary they recommend prioritizing for this adjustment? Or is that entirely up to the team's discretion?

Russel Walters: Well, it's certainly up to the team's discretion. I guess one thing I'll point out within the dataset, there's an offensive line and a defensive pairing. The defensive pairings metrics are sort of the, the opposite of the opponent offensive line metrics. So, for the home team's defense, if the away team has a high XG, or scores a lot of goals, that's indicative of a poor defense on the home team, and vice versa. So, if the home team's defense does not allow many goals, or a low XG for the away team's offense, and that's indicative of a Quality defense for the home team?

But there are no additional, defensive line-specific measures.

Liz Beard: Yeah, that's definitely something that you and your team should think about the best way to derive those metrics. And I also will use this as an opportunity to plug our educational modules again, so check those out as well for more context into how World Hockey League is run, also for how these types of factors might contribute to things like 1B, like that offensive line disparity.

Jamie Powers: Alright, the next one is kind of admin.

One of my team members had to step down due to time constraints. Can I replace them with another teammate?

Liz Beard: Awesome, great question. These things happen. They happen more often than one might think, but, as long as they are not the student team leader who is responsible for submitting answers, at the end of that Phase 1 submission form, where you'll submit all of your work and your answers, you'll be asked to confirm your team composition. So, if anyone has changed, or if you've dropped a member of your team, you can just not list them this time or list a different student. And then that's the team that we'll use moving forward for evaluating and inviting into the semi-finals.

Jamie Powers: So the next question says, for the 16 tournament matchups, you asked for the win probability of the home team. Is there a calculated home ice advantage built into the simulation, or is the home team simply the higher seed playing at a neutral site?

Russel Walters: We did talk about Home Ice Advantage in the educational modules. There is a Home Ice Advantage, I'm not gonna tell you what that is exactly, that's something that is in the simulation.

Jamie Powers: Alright, the next question is about grading. So, it asks, is AI used to grade our responses? If so, what part is it grading?

Russel Walters: AI is used. We do a lot of work in AI, it's used extensively in developing lots of things here. It's not going to be used to grade your win probability.

Liz Beard: I'll also add that, yeah, obviously AI is new and getting better all the time, but in years past, we have not used AI to grade responses. We have used some kind of natural language processing to do our own internal research about the competition, so learning about the different types of methodologies that your teams used. That being said, if we use it in any way this year,I think I can say that we will disclose that in the playbook, but in terms of actually determining semi-finalists, there's not going to be any AI involvement.

Jamie Powers: That's all the questions that we had come in during the webinar. If anybody else has anything, I would drop it in now. We got a few extra minutes.

If you don't have anything right now, but something comes up, you can always submit it to the question form on the website. We are happy to answer it there as well.

But, these were really great questions, and I feel like the longer we sat on here the more questions were coming in, it really got us thinking about the competition. We are really excited to see your Phase 1 submissions and see what you guys all came up with. So, we really wish you the best of luck. And I think that covers everything.

Liz Beard: Yeah, thank you so much, everyone, for your questions. Jamie, do you also want to talk about the transcripts that'll go out?

Jamie Powers: Yes, we are going to transcribe this video, we'll send it out to everybody, so if you couldn't make it to the webinar time, you guys will still get your answers. That should be out next week. So, you guys can go back and read through it.

Liz Beard: And that'll be coming out to you over email, so you'll get an email from our SurveyMonkey Apply site.

Jamie Powers: Alright, and if you guys are in the spirit, I hear there's a good hockey game for the Winter Olympics coming up today!

Alright, thank you everybody, I hope you have a great rest of your day.

Liz Beard: Thanks so much!

Russel Walters: Good luck.